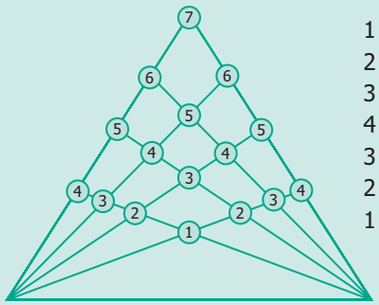
The place in which the things are counted is fixed and arranged in some order, then counting is easy, otherwise it is difficult.

This suggests that the things should be in order if it is needed to be counted easily.

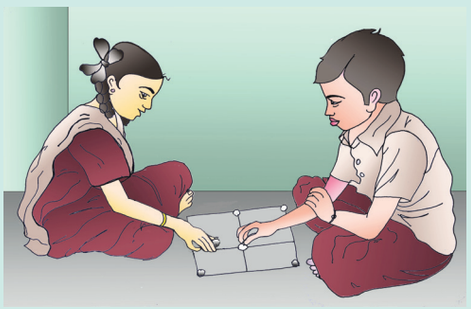
MATHEMATICS ALIVE - INFORMATION PROCESSING IN REAL LIFE



$1 \times 1 = 1$
$2 \times 2 = 4$
$3 \times 3 = 9$
$4 \times 4 = 16$
$3 \times 5 = 15$
$2 \times 6 = 12$
$1 \times 7 = 7$
64

Systematic sorting of triangles

Fig.6.4



Logical reasoning in 3 kal-aattam

Fig.6.5

6.2 Systematic listing

There are two shorts and three shirts. In how many different combinations can you wear them?

At a time you are going to pick one of the shorts and different shirts. We denote the shorts as A and B, and the shirts as p, q and r. Then you can wear one of these: A p, A q, A r, B p, B q and B r.

Thus you have six different combinations and you know that you have counted all the possibilities. This is systematic listing.

Situation 1

Your friend has built a house with three floors. He wants to paint each floor with three different colours red, blue and green. Can you help him to find different ways of possible colour combinations to paint the home?

In how many ways can his house be painted with these three colours?

Let us consider the floors as the Third floor, the Second floor and the First floor. We can say that



Fig.6.6

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